16 Express $\sqrt{45} + \frac{45}{\sqrt{5}}$ in the form $a\sqrt{5}$ where a is an integer.

$\sqrt{45} = \sqrt{(9 \times 5)} = 3\sqrt{5}$

1 mark
Simplify the surd.

$\frac{45}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{45\sqrt{5}}{5} = 9\sqrt{5}$

1 mark
Rationalise.

$3\sqrt{5} + 9\sqrt{5} = 12\sqrt{5}$

1 mark
a = 12.

.....

(Total for Question 16 is 3 marks)

DO NOT WRITE IN THIS AREA